



Robust process for high yield conversion of non-degradable polyethylene to a biodegradable plastic using a chemo-biotechnological approach[☆]

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ABSTRACT

In this study, the optimisation of a process for producing medium-chain-length polyhydroxyalkanoate (mcl-PHA) by *Pseudomonas putida* KT2440 when fed with a polyethylene (PE)-derived fatty acid mixture was investigated. The PE was pyrolysed to produce a hydrocarbon wax that was subsequently oxidised to produce a mixture of fatty acids, purified, and used as a PHA substrate for the growth and selection of microorganisms. Based on the shaken flask screening, a production strain, i.e., *Pseudomonas putida* KT2440, was selected for conducting bioreactor studies. Feeding PE-derived fatty acids in a 20-L setup resulted in high mcl-PHA yields (83.0 g L⁻¹ CDW with 65% PHA in 25 h). Furthermore, life-cycle assessment (LCA) was conducted to determine the environmental advantages of the proposed process and its impacts compared to those of other technologies for treating PE-derived waste streams. We conclude that processing waste PE into PHA, rather than incineration, produces biodegradable material while also reducing the additional emissions that arise from traditional PE waste treatment processes, such as incineration to gain energy.

1. Introduction

The circular economy seeks to move society towards increased efficiency in production and preventing the generation of waste, such as plastic (European Parliament and Council, 2008; Wojnowska-Baryła et al., 2020). While ambitious, post-consumer plastic recycling targets have been set, the recycling of plastics is limited by the quality and price of recycled polymers relative to those of virgin materials. New technologies for converting plastics into high-quality products with new functionality, such as biodegradability, are required to drive the recycling of plastics away from directly competing with virgin materials (Al-Salem et al., 2009; Panda et al., 2010). The bioeconomy is coincident with the circular economy, and seeks to replace fossil fuels with bio-

based resources. The bioeconomy initiative includes the direct replacement of major petrochemical polymers, such as polyethylene terephthalic acid (PET) and polyethylene, with their bio-based equivalents (Cristóbal et al., 2016; Tharanathan, 2003; Innovating for sustainable growth. A bioeconomy for Europe, 2012). Although such solutions address the start-of-life resources, they do not address the post-consumer end-of-life management of the plastic. Bio-based plastics such as Bio-PE are hardly biodegradable and need to be recycled post-use. Hence, preliminary reports exist on PE partial decomposition by microbiota, more robust technologies need to be developed for its practical biological removal (Montazer et al., 2020).

Previously, we reported a chemo-biotechnology processes that enable the conversion of polystyrene, PET, and PE to medium chain

[☆] We dedicate this manuscript to our friend Dr Shane T. Kenny, who sadly left us too early.

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length polyhydroxyalkanoates (mcl-PHA) (Goff et al., 2007; Kenny et al., 2008; Guzik et al., 2014). Mcl-PHA is a partially crystalline biodegradable and biocompatible polymer with a wide range of applications, from packaging to medical use (Guzik et al., 2020). The first step of its production process is the pyrolysis of plastic to generate hydrocarbon break-down products (monomers), which can act as growth and carbon substrates for PHA accumulation (Guzik et al., 2014). The hydrocarbon mixture produced by polyethylene pyrolysis is highly insoluble in aqueous media, thus limiting bacterial growth and PHA accumulation (Guzik et al., 2014). To improve the efficiency of the technology for transforming PE into PHA, the conversion of PE-derived monomers to mcl-PHA must be improved. In this study, we oxidised a PE-derived hydrocarbon mixture into a mixture of fatty acids, which are much more bioavailable for the growth of bacteria and subsequent accumulation of PHA, to alter the bioavailability of polyethylene breakdown products. Radecka et al. also attempted to improve scl-PHA (PHB) production by growing *Cupriavidus necator* H16 directly on raw oxidised polyethylene to produce PHB in shaken flasks with volumetric productivities of 0.076 and 0.026 g L⁻¹h⁻¹ of CDW and PHA, respectively (Radecka et al., 2016).

Here, we present a proof-of-concept study for the oxidation of PE-derived hydrocarbons into a mixture of fatty acids that was successfully converted into a medium chain length polyhydroxyalkanoate polymer in a 20-L bioreactor. This study describes a preliminary screening of strains that can grow and produce mcl-PHAs from a synthetic mixture of fatty acids and serves as a benchmark for development work. Furthermore, an adapted fermentation feed profile enabled the growth of a *P. putida* KT2400 culture with a high content of mcl-PHA polymers to a high cell density at a 20-L scale. The polymer was thoroughly characterised following standard polymer science methods (differential scanning calorimetry, DSC; thermogravimetric analysis, TGA; X-ray diffraction, XRD, gas chromatography, GC; and gel permeation chromatography, GPC).

To provide our development with environmental context, we also conducted a life cycle assessment that compared three different scenarios with the post-consumer polyethylene as the point of origin. The first, second, and third scenarios considered: (1) the burning of PE wax for energy generation; (2) the use of PE wax alone for PHA production as per our previous study (Guzik et al., 2014); and (3) the process described herein that enables PHA production from PE-derived fatty acids, respectively. There were significant differences between the outputs of scenarios 1 and 2 and 3. Scenario 1 leads to energy recovery, while scenarios 2 and 3 produce PHA. As the examination was based on LCA methodology, the ISO 14040/14044 guidelines were used as a reference. The environmental impact was calculated following the ReCiPe and SimaPro software. The flows within the analysed life cycles were modelled using primary data from experimental processes and secondary data from different sources, including the Ecoinvent v3 database, mathematical modelling, and literature.

2. Materials and methods

2.1. Materials

2.1.1. Bacterial growth medium

The minimal mineral salts medium (MSM) contained the following per litre: 4.70 g (NH₄)₂SO₄, 3.00 g MgSO₄·7H₂O, 15.00 g Na₂HPO₄·7H₂O, 3.40 g KH₂PO₄, and trace elements as defined by (Sun et al., 2007a), and was used for all flask cultivations and fermentations conducted in this study. A modified MSM medium (MSM/MOPS) was prepared to limit the amount of phosphorus, in which all of the phosphorous sources were exchanged with 50 mM of 3-(N-morpholino) propanesulfonic acid (MOPS) buffered with KOH to pH 6.9. To prepare solid media, 15 g L⁻¹ of agar was added to the MSM.

2.1.2. Bacterial strains used

The ability of a selection of known bacterial strains to grow on and produce PHA from a range of substrates was tested (Table S1). The strains were chosen based on previous reports that provided information on growth and PHA production from fatty acids by these strains.

2.1.3. Substrates

A synthetic fatty acid mixture (SFAM) ranging from butyric to tetradecanoic acid, which resembles the predicted product of the oxidation of the PE pyrolysis product, was supplied as the carbon and energy source for all feed development experiments (Figure S1).

Oxywax Ltd. conducted the oxidation of a polyethylene-derived hydrocarbon mixture (PE wax) (Guzik et al., 2014) using an in-house technology that reacts a hydrocarbon mist with air at an elevated temperature (–100–140 °C) in the presence of a permanganate catalyst until a specific acid value is reached. The process generates carboxylic acids by paraffin carbon chain scission and unreacted hydrocarbons. The PE wax sample exhibited a green gelled appearance at 10 °C with a pronounced odour of thermos-cracked fluids. The sample was transferred to a stainless-steel vessel fitted with a bubble-forming sparger, Silverson mixer, and condenser, and the oxidation process was conducted at atmospheric pressure. Oxygen was introduced using an oxygen generator at a rate of 1 L min⁻¹. After heating the sample to 110 °C, the Mn catalyst (1.0 g of catalyst per 0.5 kg of sample) and 50% w/w KOH (1.0 mL of solution per 3.0 kg of sample) were added while mixing the sample. Strong, exothermic conditions were maintained between temperatures of 132 and 140 °C. The sample volatility was significant; therefore, after 4 h of oxidation, the reaction was terminated after reaching an acid no. of 25 mg KOH/g.

The fatty acid fraction was recovered from the remaining unreacted hydrocarbons following a modified Bocca method based on selective solvent extraction (Bocca et al., 1971). Briefly, a crude mixture was treated with *N,N*-dimethylformamide (DMF) at a ratio of 1:2 and 55 °C. The upper phase, consisting of insoluble hydrocarbons, was separated from the lower phase, consisting of the DMF-soluble fraction. The solution was cooled to 25 °C and washed twice with *n*-hexane. The DMF was removed using a rotatory evaporator at 30 mbar and 100 °C. The resulting polyethylene-derived fatty acid mixture (PE-FAM, Fig. 3) was then used as a substrate in the process validation experiments to demonstrate the technology for transforming PE into PHA.

2.2. Biological and analytical studies

2.2.1. Growth conditions for PHA accumulation in the shaken flask experiments

All strains were grown in 250-mL Erlenmeyer flasks containing 50 mL of nitrogen-limited MSM medium and various carbon sources with a final concentration of 1.95 of carbon per litre (g_C L⁻¹). The flasks were incubated at 30 °C and shaken at 250 rpm for 48 h. To screen for organisms that can accumulate PHA, the content of the inorganic nitrogen source, i.e., ammonium chloride (NH₄Cl), was limited to 0.25 g L⁻¹ (65 mg N L⁻¹ = 4.6 mM). For phosphorous-limited growth, the MSM was exchanged with an MSM/MOPS liquid medium using the only phosphorous source from the 100-μL overnight inoculum (grown in normal MSM), which supplemented the MSM/MOPS with up to 22 mg P L⁻¹ (7.2 mM).

2.2.2. Inoculation culture conditions for bioreactor studies

A *P. putida* KT2440 culture was prepared by inoculating a clean *Pseudomonas* isolation agar (PIA) plate with a single colony obtained from colonised PIA. The culture was incubated at 30 °C for 24 h. Then, 50 mL of MSM broth supplemented with SFAM containing 1.95 g g_C L⁻¹ was inoculated with a loop of this culture and incubated at 250 rpm and 30 °C for 12–16 h.

2.2.3. Fed-batch fermentation conditions of the bioreactors

Fed-batch fermentation was conducted in a Biostat C+ fermenter (Sartorius AG, Germany) with a working volume of 20 L. Oxygen was supplied in the form of air (not supplemented with pure oxygen). The initial carbon source (PE-FAM) concentration for the initial batch period was 0.84 gC L^{-1} . The initial Biostat C+ fermentation (20 L) conditions were as follows: starting volume, 12 L; agitation, 300 rpm; aeration, from 0 to 0.3 h, no air was added due to rapid foam formation; thereafter, air was added at 6 L min^{-1} ; dissolved oxygen (DO) – controlled at 20% by increasing the agitation and airflow rates up to 30 L min^{-1} ; pH – controlled automatically at 6.9 ± 0.1 by adding 20% NH_4OH (Sigma) or 15% v/v H_2SO_4 (Sigma); and foaming was controlled by the automatic addition of antifoam (polypropylene glycol P2000, Sigma) when required. The DO content was controlled at a lower level (20%) in the 20-L fermentation device. Data from each fermentation experiment were recorded using the BioPAT® MFCS SCADA fermentation software (Sartorius AG, Germany).

2.2.4. Nutrient and biomass analysis

Samples were collected at regular intervals during each fermentation cycle to analyse the concentrations of fatty acids, inorganic nutrients, and biomass. The nitrogen concentration in the growth medium was monitored over time following the method described in (Scheiner, 1976). Additionally, the phosphate concentration was determined following the US EPA colourimetric method (USEPA, 1978). Fatty acids were extracted from the culture media and analysed following the method described in (Lalman and Bagley, 2004).

2.2.5. Downstream processing

The fermentation broth was centrifuged using a continuous flow centrifuge (Heraeus Contifuge Stratos, Thermo Fisher Scientific, Germany) at 4°C and 25,040g in a titanium rotor (HCT22.300, Thermo Fisher Scientific, Germany). Cells were first frozen at -20°C for 4–6 h, transferred to a -80°C freezer overnight, and then lyophilised using a freeze-dryer system (FreeZone bulk tray dryer, Labconco, USA) until they were fully dried. PHA was extracted and precipitated following the procedure described in (Elbahloul and Steinbüchel, 2009), in which concentrated PHA was precipitated in a 70% v/v ethanol and methanol ice-cold water solution with an alcohols ratio of 1:1.

2.2.6. PHA content and monomer composition of the bacterial cultures

The polymer content of lyophilised whole bacterial cells was determined by subjecting the cells to acidic methanolysis following previously described protocols (Brandl et al., 1988). The (R)-3-hydroxyalkanoic acid methyl esters were assayed by gas chromatography (GC) using an Agilent 6890 N chromatograph equipped with a BP21 capillary column ($25 \text{ m} \times 0.25 \text{ mm} \times 0.32 \mu\text{m}$; SGE Analytical Sciences) and flame ionisation detector (FID) with the following temperature program: 120°C for 5 min, followed by an increase of 3°C per min until 180°C , which was maintained for 10 min. Commercially available (R)-3-hydroxyalkanoic acids were methylated as described above to identify the peaks of the PHA samples (Bioplastech Ltd., Dublin, Ireland). The PHA monomers were identified using an Agilent 6890 N GC fitted with a 5973-series inert mass spectrophotometer (MS) and HP-5 column ($12 \text{ m} \times 0.2 \text{ mm} \times 0.33 \mu\text{m}$; Hewlett-Packard) was used with the following temperature program: 50°C for 3 min, followed by an increase of 10°C/min to 250°C , which was held for 1 min.

2.2.7. PHA polymer and analysis

Differential scanning calorimetry (DSC) was conducted using a Perkin Elmer Pyris-Diamond Calorimeter calibrated using indium standards. The samples weighing 7–8 mg were encapsulated in hermetically sealed aluminium pans and heated from -70 to 100°C at a rate of $10^\circ\text{C min}^{-1}$. To determine the glass transition temperature (T_g), the samples were maintained at 100°C for 1 min and rapidly quenched to -70°C , and then re-heated from 100°C at $10^\circ\text{C min}^{-1}$ to determine the melting

temperature (T_m) and T_g . The T_m was measured at the peak of the melting endotherm, while the T_g was taken as the midpoint of the heat capacity change.

Thermogravimetric analysis (TGA) was conducted on a Perkin Elmer Pyris 1 thermogravimetric analyser calibrated using nickel and iron standards to determine the thermal stability and decomposition profile of the samples. Approximately 7 mg of each sample was placed in a platinum pan and heated from 30 to 700°C at a heating rate of $10^\circ\text{C min}^{-1}$ under an air atmosphere.

The molecular weight distributions were determined by gel permeation chromatography (GPC) using a 5-mm PL mixed-C + PL gel column (Perkin Elmer) with a PELV 290 UV-Vis detector set at 254 nm . Spectroscopic-grade tetrahydrofuran (THF, Sigma) was used as the eluent at a flow rate of 1.0 mL min^{-1} . The sample concentration and injection volume were set at 1% and $500 \mu\text{L}$, respectively. A molecular weight calibration curve was generated from polystyrene standards with low polydispersity using Turbochrom 4.0.

X-ray diffraction (XRD) was conducted at room temperature ($\sim 20^\circ\text{C}$) and the diffraction patterns were collected on a Siemens D500 diffractometer fitted with a $\text{Cu-K}\alpha$ radiation source ($\lambda = 0.1514 \text{ nm}$) operated at 40 kV and 30 mA . Data were obtained from 2 to 60° (2θ) at a scanning speed of $0.1^\circ \text{ min}^{-1}$. XRD allowed to detriment a degree of crystallisation of the synthesised PHA polymer.

2.3. Life-cycle assessment

2.3.1. Goal and scope of LCA

The LCA study mainly aimed to compare three different waste PE end-of-life scenarios describing the possible flows of PE end-of-life phases. Table 1 presents the general characteristics of the scenarios, focusing on the processes, by-products, and products. The avoided products are also listed for each of the scenarios, representing the possibility of replacing raw products with their waste PE-based alternatives. Scenario 1 is a commonly recognised and practised process, while Scenarios 2 and 3 are currently experimental. Additionally, the potential of the commercialisation of the investigated processes was also estimated.

The functional unit of the study was 1 kg of post-consumer PE processed in one of the included scenarios. This study focused on end-of-life processes; therefore, the preceding phases in the life cycle were only summarised with the waste PE input. The definition of a functional unit as an input unit rather than an output unit is uncommon in LCA studies, as they focus on products. However, this is common when considering waste treatment and processing scenarios, where different alternatives of processing end with different outputs are considered.

The system boundaries were defined to only include part of the life cycle focused on waste PE treatment (Fig. 1). The geographical extent was set to production in the EU.

2.3.2. Life-cycle inventory

Data regarding the investigated scenarios of waste PE treatment were obtained from several sources. The key source for modelling scenarios 2 and 3 was the primary data from experimental PHA production presented in (Guzik et al., 2014) and this study. The raw data regarding the rhamnolipid fermentation process used in Scenario 2 were also

Table 1
End-of-life scenarios for waste PE.

Scenario No.	Input	End-of-life processes	By-products and products	Avoided product
Scenario 1	Waste PE	Pyrolysis Incineration	Mixtures of alkanes; Energy; CO_2 emissions	Energy
Scenario 2	Waste PE	Pyrolysis Fermentation	Mixtures of alkanes; PHA	PHA
Scenario 3	Waste PE	Pyrolysis Oxidation Fermentation	Mixtures of alkanes; Fatty acids; PHA	PHA

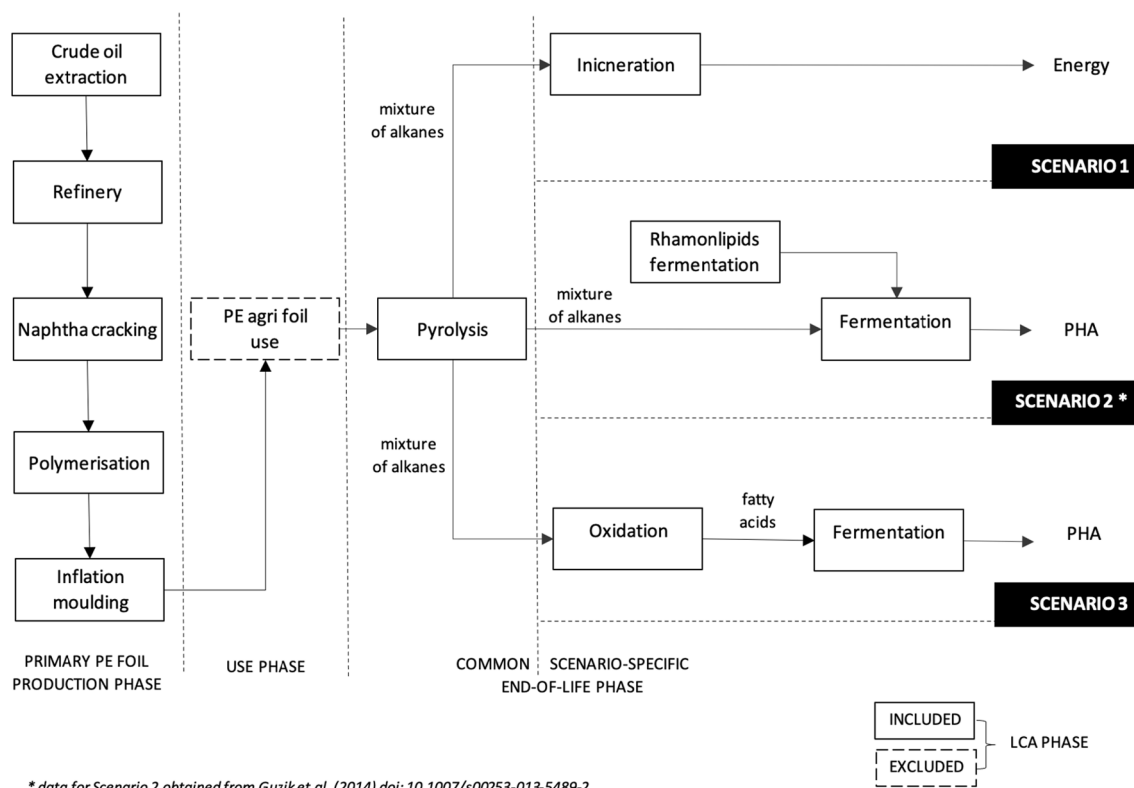


Fig. 1. System boundaries for the envisaged scenarios.

supplemented with secondary data (Kopsahelis et al., 2018). The bacterial fermentation process was based on raw data and was consistent with that presented in (Nitkiewicz et al., 2020). Scenarios 2 and 3 were based on primary data, and the pyrolysis processes were based on secondary data from literature (Ricardo-AEA, 2013; Ahmad et al., 2015; Sogancioglu et al., 2017). The raw data regarding the incineration process in Scenario 3 were combined with data from the Ecoinvent v3 database. Data regarding the energy used for operating laboratory equipment during experiments under Scenarios 2 and 3, including stirrers, ovens, lipid extractors, rotary evaporators, shaker tables, and centrifuges, were estimated based on raw operating time and load data, and secondary average energy use data (Piccinno et al., 2016; Szepessy and Thorwid, 2018; Evans, 2019). All of the remaining data were obtained from the Ecoinvent v3 database.

2.3.3. Life-cycle impact assessment

This study aimed to provide a general overview of the impacts related to different PE recycling scenarios following the ReCiPe impact assessment method, which has good coverage for EU-based production processes and covers all possible impacts. This method was equipped in midpoint and endpoint sets of indicators, and allows problems related to the calculation and interpretation of results to be avoided, and switching between the two options to gain better coverage of the investigated issues (Goedkoop et al., 2013). The ReCiPe method calculates 18 midpoint-level and 16 endpoint-level indicators, which are accumulated in three damage-category indicators. The 18 midpoint-level indicators included: terrestrial acidification, climate change, fossil depletion, metal depletion, water depletion, freshwater ecotoxicity, marine ecotoxicity, terrestrial ecotoxicity, freshwater eutrophication, marine eutrophication, human toxicity, ionising radiation, agricultural land occupation, natural land transformation, urban land occupation, ozone depletion, particulate matter formation, and photochemical oxidant formation. Water depletion and marine eutrophication were excluded from the endpoint set, while the damage categories were grouped into three

protection areas, i.e., human health, resources, and ecosystems (Huijbregts et al., 2017).

The indicator in this study was ReCiPe Endpoint (H) V1.08, referring to Europe, with the hierarchist perspective used and allocation mechanisms and weighting scheme proposed by its authors. SimaPro 8.5.2.0 with Ecoinvent 3.1 were used for all calculations.

2.3.4. Cost estimation on PE recycling scenarios

Since the process presented in the paper is not implement in industrial scale yet its cost could be only estimated basing on actual prizes of energy and supplies needed for the processing. In order to estimate the costs of the investigated scenarios the following assumptions are made: i) only the scenario specific extra costs are considered (estimation does not cover all the process); ii) estimation is based on world average prices of energy in 2019 for industrial sector as calculated by IEA (<https://www.iea.org/reports/energy-prices-overview>) and on actual average prices of chemical supplies as offered by vendors for industrial purposes (for details consult Table S2); iii) impact of scaling-up factors is not considered in the estimation and all the calculations are made for functional unit of the study, which is 1 kg of waste PE.

3. Results and discussion

While we awaited the results of PE wax oxidation by Oxywax Ltd., we conducted preliminary tests of the abilities of known PHA accumulators to produce biopolymers from SFAM based on the predicted oxidation product of PE wax (Bednarska et al., 1987). The theoretical content of fatty acids that would have been generated by the method reported by Bednarska et al. was calculated (Figure S1) based on the hydrocarbon content of the PE-derived wax (Figure S2). This SFAM was administered as sodium salts in subsequent flask experiments.

Screening of a variety of strains in shaken flasks revealed that *P. putida* GO16 achieved the highest CDW and accumulated high levels of PHA in shaken flask experiments (Table S3). However, preliminary

fermentation experiments indicated that there was no difference between *P. putida* GO16 and *P. putida* KT2440 (data not shown). Therefore, detailed studies regarding the utilisation of the synthetic fatty acid mixture over time were conducted using *P. putida* KT2440, as it is a much better studied strain and a well-known PHA accumulator that can accumulate high PHA levels from fatty acids in bioreactors (Lee et al., 2000; Maclean et al., 2008). After a lag period of approximately 6 h, *P. putida* KT2440 utilised each of the fatty acids from the mixture at an average rate of $0.015 \text{ g}_C \text{ l}^{-1} \text{ h}^{-1}$ over the following 6 h (Fig. 2; for the individual fatty acid utilisation rates within the mixture please consult Table S4).

3.1. Conversion of the PE-derived fatty acid mixture to PHA in shaken flasks by *P. putida* KT2440

The PE pyrolysis product was converted into a mixture of fatty acids by chemical oxidation. The oxidation of PE reached 35% completion, and the resulting product was a mixture of fatty acids and unreacted hydrocarbons. *P. putida* KT2440 could not grow on this raw product; therefore, the mixture underwent separation using DMF extraction, which resulted in near total separation of fatty acids from the other components of the raw product as evidenced by GC-FID analysis. The composition and distribution of the fatty acids of the DMF extracted mixture (PE-FAM) consisted of C_3 to C_{17} fatty acids (Table S5).

The PE-FAM was added to shaken flask growth experiments in nitrogen-limited MSM medium with *P. putida* KT2440. Over a 48-h incubation period, the strain achieved $0.43 \text{ g L}^{-1} \text{ CDW}$; however, PHA accumulation was not observed, even though all of the detectable inorganic nitrogen was utilised within the first 30 h. An additional nitrogen source may have been introduced into the culture due to the clean-up procedure employing DMF as the primary solvent. It was previously reported that *Pseudomonads* can utilise DMF as a nitrogen source (Schar et al., 1986). Therefore, we limited different inorganic nutrients in the culture media to promote PHA accumulation, including phosphorous. The amount of phosphorous introduced under these conditions (maximum ratio of C:P of 8.7) was sufficient to allow growth, but also promoted PHA accumulation. *P. putida* KT2440 achieved $0.17 \text{ g L}^{-1} \text{ CDW}$ under these conditions, with 25.2% of the CDW as PHA. The growth yield on this substrate ($Y_{CDW/S} = 0.22 \text{ g}_C \text{ g}_C^{-1}$) was identical to

that obtained from SFAM ($Y_{CDW/S} = 0.22 \text{ g}_C \text{ g}_C^{-1}$) for this strain when low nitrogen containing MSM was used. The conversion of PE wax to a mixture of fatty acids resulted in a 1.4-fold improvement in biomass accumulation compared to the growth of *P. aeruginosa* GL-1 supplemented with PE wax in the presence of rhamnolipids (Guzik et al., 2014). When phosphorous was limited, the PHA accumulation improved 2.0-fold when compared to *P. aeruginosa* GL-1 grown on PE wax. Additionally, as *P. aeruginosa* GL-1 is an opportunistic pathogen, it is beneficial to use a non-pathogenic strain, such as *P. putida* KT2440, in industrial processes. Additionally, there were no significant differences in the compositions of the monomers accumulated by *P. putida* KT2440 from PE-FAM and SFAM (Figure S3).

3.2. Conversion of PE-FAM to PHA using *P. putida* KT2440 in a 20-L bioreactor

PHA fermentation processes require growth to a high cell density as a basis for high PHA productivity and fed-batch fermentation is one of the most efficient methods of achieving this (Lee, 1996; Lee et al., 1999; Sun et al., 2007b; Elbahloul and Steinbüchel, 2009). In several studies applying high-cell-density cultures to mcl-PHA production, both the mcl-PHA and biomass levels were improved when compared to flask-scale experiments (Kellerhals et al., 1999; Cardoso Diniz et al., 2004; Sun et al., 2009; Scheel et al., 2021). We used a combination of two fermentation feeding strategies established in (Maclean et al., 2008) and (Lee et al., 2000) to grow *P. putida* KT2440 on PE derived fatty acids in a bioreactor. Maclean et al. used nonanoic acid as the feed substrate for the growth of *P. putida* KT2440 and accumulation of mcl-PHA ($109 \text{ g L}^{-1} \text{ CDW}$ with 63% PHA), while Lee et al. used a phosphorus-limited regime when growing *P. putida* KT2440 with oleic acid as the sole carbon and energy source achieving a CDW of $141 \text{ g L}^{-1} \text{ CDW}$ with 51.4% PHA. We adjusted the profile of the substrate feeding strategy process proposed by Maclean et al. to allow efficient use of PE-FAM avoiding accumulation of fatty acids in the medium. Secondly, based on the preliminary bioreactor experiments (data not shown), we developed a cultivation medium to ensure that, at 20 h, the supplemented phosphate salts became the limiting growth factor. This hybrid feeding profile for the fermentation of PE-FAM at a 20-L scale was successful and resulted in $83.0 \text{ g L}^{-1} \text{ CDW}$ with 59% PHA within 25 h (Fig. 4). This resulted in an overall biomass

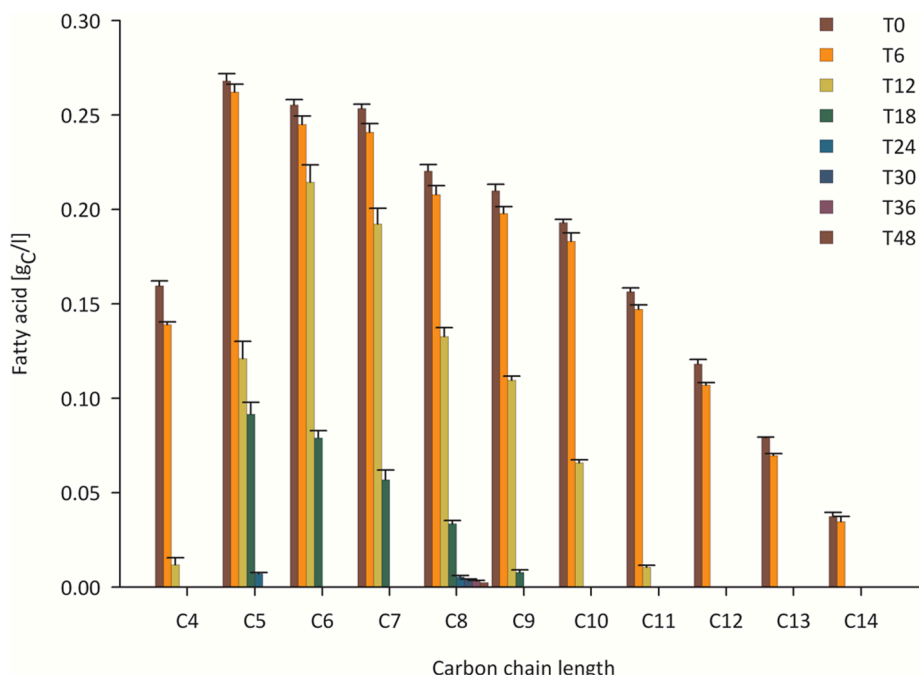


Fig. 2. SFAM utilisation by *P. putida* KT2440 over a 48-h period. The initial amount of carbon supplied was $1.95 \text{ g}_C \text{ l}^{-1}$.

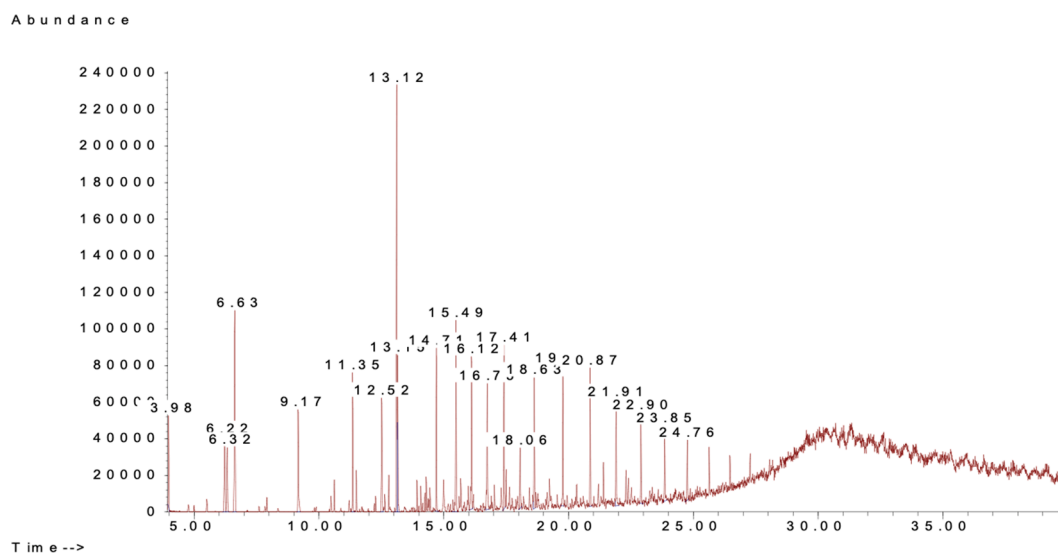


Fig. 3. Polyethylene derived fatty acid mixture (PE-FAM) composition after extraction with DMF. A detailed composition of this substrate is presented in Table S5.

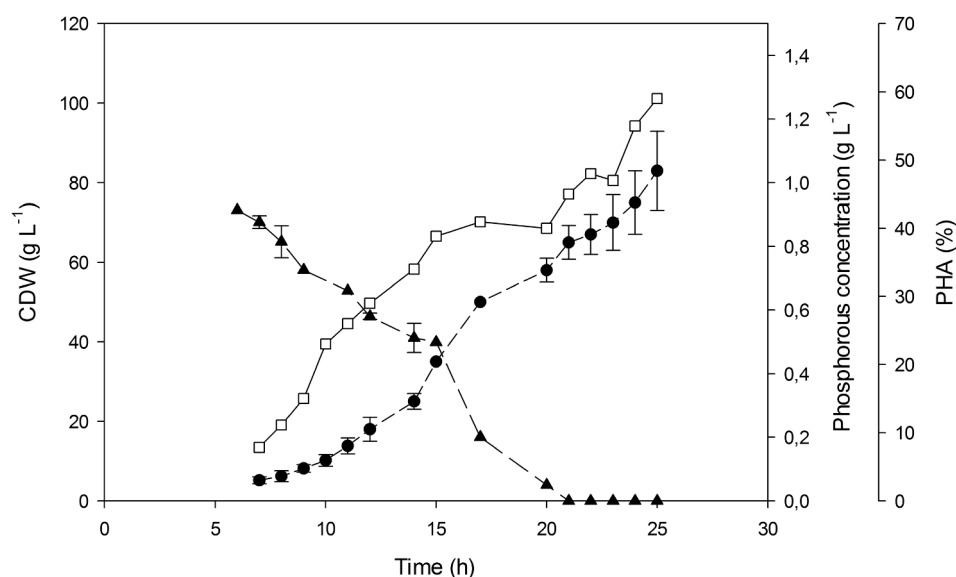


Fig. 4. Growth of *P. putida* KT2440 on PE-FAM fed following a novel feeding strategy at a 20-L scale with inorganic phosphate limitation from hour 20 in a Biostat C + fermenter. The cell dry weight (CDW) (g L⁻¹, ●), PHA concentration (%), and P concentration (g L⁻¹, ▲) were tracked over time.

volumetric productivity and corresponding PHA volumetric productivity of 3.3 and 2.1 g L⁻¹h⁻¹, respectively.

The (*R*)-3-hydroxyalkanoic acid composition of the extracted mcl-PHA polymer was very similar to that obtained by cultivating *P. putida* KT2440 in shaken flasks. The polymer contained both odd and even saturated (*R*)-3-hydroxyalkanoic acid monomer units. The predominant monomers were (*R*)-3-hydroxyoctanoate (average. 27 mol%) and (*R*)-3-hydroxynonanoate (average. 24 mol%), followed by (*R*)-3-hydroxyheptanoate (average. 17 mol%) and (*R*)-3-hydroxydecanoate (average. 16 mol%). The yields of the remaining monomers were below 10 mol% each, and included (*R*)-3-hydroxyhexanoate (average. 7 mol%), (*R*)-3-hydroxyundecanoate (average. 6 mol%), (*R*)-3-hydroxy-dodecanoate (average. 3 mol%), and (*R*)-3-hydroxytridecanoate (average. 1 mol%).

Thermal degradation onset for the polymer occurred at approximately 291 °C, resulting in partial crystallinity, as indicated by the presence of a melting peak. The melting temperature of PHA produced from PE-FAM was 34.9 °C. Thermal analysis indicated a low glass transition temperature of approximately -43.1 °C. The ΔH value was

10.0 J g⁻¹, indicating that the polymer was predominantly amorphous. The molecular weight of the mcl-PHA isolated from cells grown under the adapted strategy was Mw = 152.9 kDa with a polydispersity index of 1.9, which is comparable to that of typical mcl-PHA elastomers produced previously (Kenny et al., 2008; Sofińska et al., 2019). These properties suggest that the PHA polymer could be applied as elastomeric additives in coatings, adhesives, as previously suggested for other similar mcl-PHA polymers (Van Der Walle et al., 1999; Guzik et al., 2020).

3.3. Life-cycle assessment of the PE upcycling technologies

Table 2 presents the LCA results for the three PE recycling scenarios. As the damage assessment results were divided by the average European contribution to the specific impact categories throughout the year, the values were unitless. Fossil depletion, climate change, and particulate matter formation were the impact categories most affected by all scenarios. Metal depletion, human toxicity, and land occupation for both

Table 2

Comparison of the LCA results for the PE recycling scenarios – normalised ReCiPe impact category endpoint indicators.

Impact category	Scenario 1	Scenario 2	Scenario 3
Climate change Human Health	0.0003	0.0013	0.0017
Ozone depletion	0	0	0
Human toxicity	0	0	0
Photochemical oxidant formation	0	0	0
Particulate matter formation	0.0001	0.0003	0.0005
Ionising radiation	0	0	0
Climate change Ecosystems	0.0002	0.0008	0.0011
Terrestrial acidification	0	0	0
Freshwater eutrophication	0	0	0
Terrestrial ecotoxicity	0	0	0
Freshwater ecotoxicity	0	0	0
Marine ecotoxicity	0	0	0
Agricultural land occupation	0	0	0
Urban land occupation	0	0	0
Natural land transformation	0	0	0
Metal depletion	0	0.0001	0.0001
Fossil depletion	0.0010	0.0025	0.0017

agricultural and urban purposes endured some minor effects. Generally, scenario 3 was dominant in all categories and has the highest cumulative impact; however, scenario 2 had a greater impact on some of the categories, including fossil and metal depletion, and agricultural land occupation. This was particularly notable for the fossil depletion impact category, which exhibited the highest overall scores and greatest difference between scenarios 2 and 3, with the best results for scenario 2.

The weighting phase of life-cycle impact assessment (LCIA) is not mandatory; however, we used it to demonstrate the overall impact of the scenarios considered here (Fig. 5, Panel A). The overall impact was highest for scenario 3, which also exhibited the greatest contribution to the human health and ecosystem damage categories. Scenario 2 had a significantly lower overall score; however, it had a predominant impact on the resource damage category.

Only scenarios 2 and 3 were included in further study due to the similarities in their outcomes, i.e., PHA production. Fig. 6 presents the weighted results of scenarios 2 and 3; however, it considers the contributions of processes to the value of the single score indicator. The contribution of each process or material is represented by the colour of the lines, with red and green indicating negative and positive environmental impacts, and their thickness, with thicker lines indicating greater impacts. For clarity, only the flows with contributions of at least 1% to the overall impact are included.

The key difference between the two scenarios considered lies within the positive impact related to the outcome of upcycling. The amount of PHA produced at the end significantly decreased the environmental impact of scenario 2, while it was almost equal to the negative impact under scenario 3. The absolute results expressed as the ReCiPe single score indicator were still higher for scenario 3; however, considering the

relative results, scenario 3 would offer a better opportunity for PE upcycling.

Electricity and heat use within the two scenarios made the greatest contributions to the environmental impact. As both processes were inventoried based on experimental processes, the critical answer would arise following scaling. The process realisation experimental set-up did not allow for the introduction of significant changes within the energy supply. However, while up-scaling the process, energy sources could be diversified, renewable energy could be introduced, or energy efficiency issues could be resolved.

To summarise the LCA results, it should be noted that the use of electricity was the feature of scenarios 2 and 3 that had the greatest impact. Introducing a new process in the form of the PE to PHA scenario significantly contributed to the increase in the impact and created doubt for the process idea. However, the arguments to investigate this further are related to the conversion efficiency of PHA from waste PE, which was over 75 times higher for scenario 3 compared to scenario 2. The streams of market-ready PHA significantly contributed to decreasing the current impact and offers even greater environmental burden mitigation with process up-scaling. The environmental impact of scenario 1 was the smallest, but only until the assessment is oriented towards waste PE input and not on recycling outcomes. If the orientation of the scenarios would be switched from the number of inputs to the number of outputs, the results would be much more favourable for scenario 3.

As the first technological process in the investigated scenarios was pyrolysis, it should be noted that it is an advantageous feature of waste PE treatment. Our study is consistent with different studies on waste PE treatment that reported that undergoing pyrolysis simplified the process by avoiding the requirement for numerous pre-treatment processes (Al-Salem et al., 2014; Ahmad et al., 2015; Antelava et al., 2019). Our study does not consider the opportunity to use waste PE with other plastics or paper in pyrolysis; therefore, we cannot discuss its potential applicability in alternative end-of-life treatments, such as landfill or gasification (Demetriou and Crossin, 2019). The end-of-life scenarios considered in this study are oriented on experimental processes under the scenario similar to the input (PE from waste Agri foil) and primary processing (pyrolysis). Future studies could focus on alternative PE waste processing methods to gain a better benchmark for waste PE-to-PHA scenarios.

The findings regarding the key role of energy use in waste PE-to-PHA when considering environmental impacts were consistent with some studies focused on laboratory-scale processes (Shah et al., 2016) and industrial processes (Benavides et al., 2017). Therefore, the common issue influencing the LCA results and its dependence on the location and local energy mix can be discussed (Laurent et al., 2014). Such discussion is important as the opportunities for energy efficiency improvement are limited in laboratory-scale processes.

An important aspect of the assessment is the impacts of specific end-of-life waste PE treatment processes. Some end-of-life scenarios have

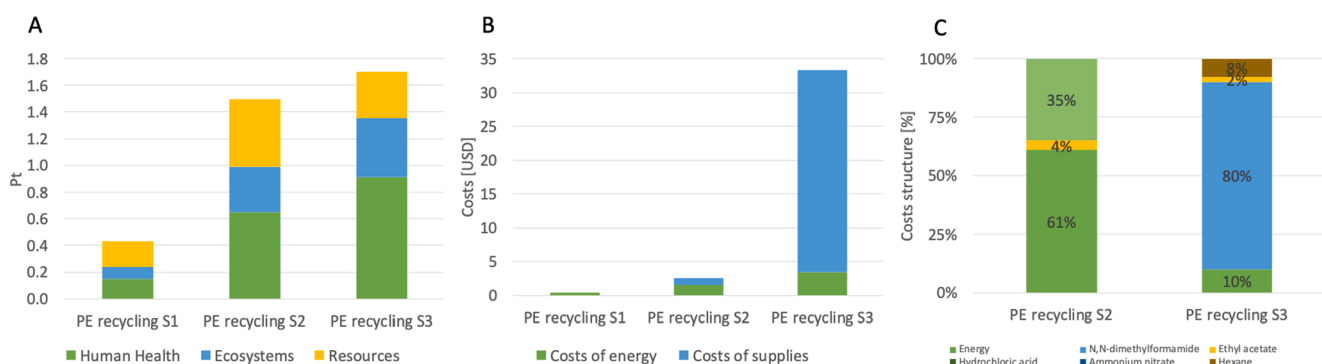
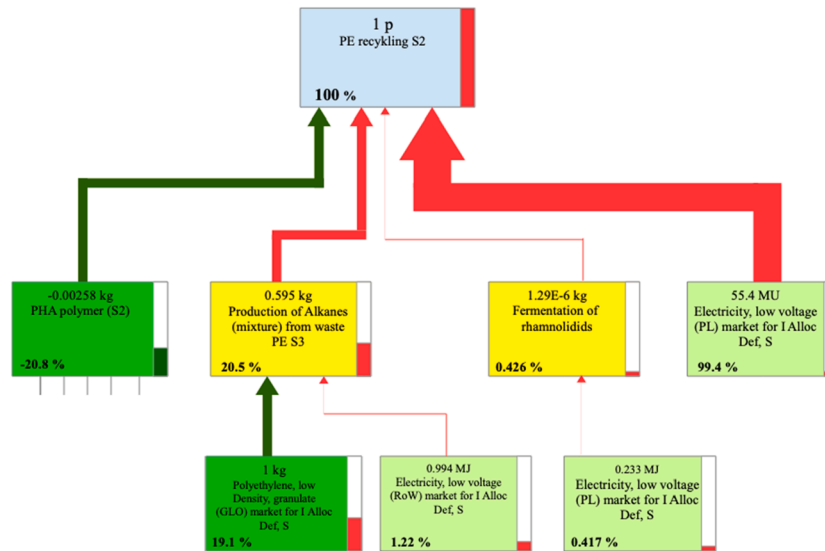


Fig. 5. Comparison of the LCA results for the PE recycling scenarios – weighted ReCiPe damage category endpoint indicators (Panel A). Cost estimation of 1 kg of PE recycling within investigated scenarios (Panel B). Comparison of cost structure between PE to PHA scenarios (Panel C).

Scenario 2



Scenario 3

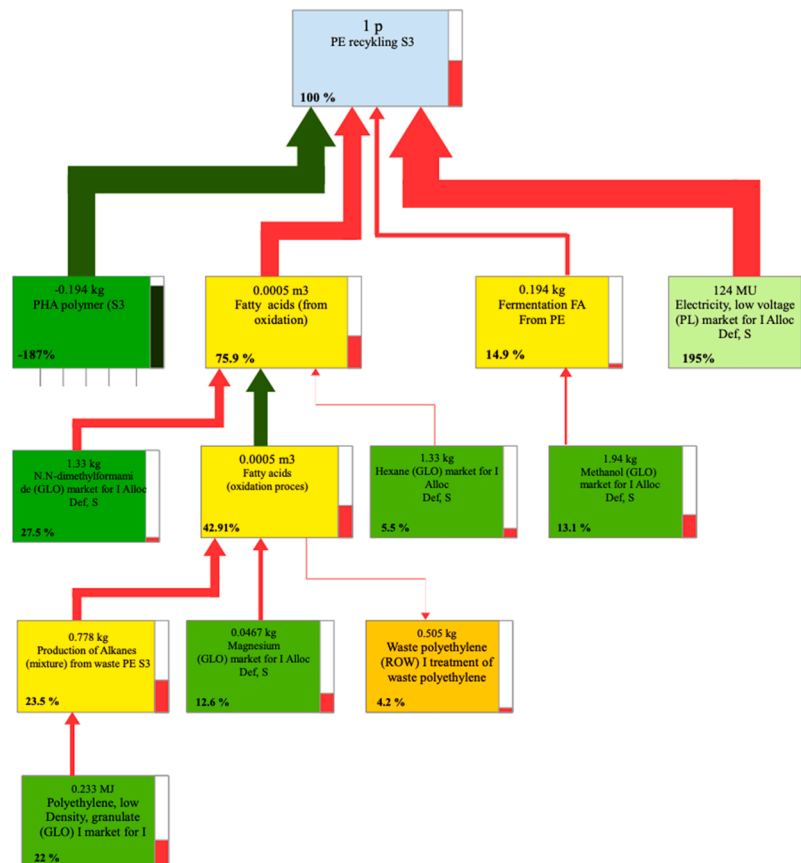


Fig. 6. Impact assessment of the different PE to PHA recycling scenarios by weighting phase. Scenario 2 – production of PHA from waste PE after fermentation; Scenario 3 – production of PHA from waste PE after oxidation and fermentation (1 % contribution cut-off).

been considered in the literature, including landfill, incineration, waste-to-fuel, mechanical recycling, or energy recovery (Antelava et al., 2019; Vogli et al., 2020); however, their setup differs to the scope of our study. PE treatment by incineration can aid in managing waste that is difficult to degrade, but produces additional emissions. Processing waste PE to PHA produces an easily degradable material, but also uses resources and

energy. The comparison presented here does not consider all of the issues, particularly the up-scaling of PHA production and possible changes in the process efficiency.

3.4. Cost estimation of PE recycling technologies

The investigation of any waste treatment or recycling technology should be also made within its economic scope. Since the technologies depicted in investigated scenarios 2 and 3 are not yet implemented within the industry, the cost assessment would be only a rough estimate of actual costs and benefits related to its implementation. Nevertheless, such an estimation is made basing on the assumptions as presented in Materials and Methods section. It is important to mention that the estimation is based on scenario specific costs only and does not include the costs that are common for all the scenarios. Fig. 5, Panel B presents the estimated costs for investigated scenarios. Since the scenario 1 is using no resources and includes pyrolysis and incineration only its cost rate is almost negligible. Despite of more complex technology used within the Scenario 2, and additional resources and energy use with rhamnolipids fermentation, its cost are also very low. In contrast to the former two scenarios, costs in scenario 3 are very high mainly due to the use of costly resources.

Fig. 5, Panel C explores further the cost structure within PE to PHA scenarios 2 and 3. Due to the use of DMF in oxidation process the structure of costs is diametrically changed in scenario 3. It is visible through total domination of DMF costs (80%) and significant decrease of energy costs share.

On the other hand, due to low gain of PHA in scenario 2 the share of costs of methanol and ethyl acetate has significantly increased. It is also worth to consider some options of economic benefits that could be produced by respected scenarios. Possible revenues from electricity sales in scenario 1 are slightly lower than the costs of energy used. Instead, in scenarios 2 and 3 the possible source of income is sales of PHA. While the scenario 3 specific costs are over 13 times higher than the costs of scenario 2, possible revenues from PHA sales in scenario 3 are around 60 times higher than those of scenario 2. Scaling up the waste PE recycling to PHA must therefore deal with the costs of oxidation process but also with the market for PHA and its available customers.

4. Conclusions

While we have reported on the conversion of polystyrene (Goff et al., 2007), polyethylene terephthalic acid (Kenny et al., 2008) and polyethene (Guzik et al., 2014) the biomass and PHA productivity in the current study is an order of magnitude higher. This conversion technology is comparable to the best reports for mcl-PHA production from virgin biological substrates, such as fatty acids (Lee et al., 2000; Maclean et al., 2008). Furthermore, our research enriches the existing knowledge regarding different PE waste management scenarios. The results indicate changes in the environmental impacts with changes in the PE waste management scenarios. Therefore, this study can help managers and policymakers understand the principles of implementing plastic waste management technologies, which are still in a relatively early stage of technical development.

The conversion of PE to PHA through a combination of chemical and biological processes fits with the vision of transitioning to a circular economy (CE), enabling the selection of solutions that promote collection of non-degradable plastics and thus prevent them entering the environment. The research conducted allows for a better understanding of the implementation of CE in manufacturing practices. Switching to CE offers an opportunity to gain a lasting competitive advantage, as this approach protects economic operators from resource scarcity and price volatility. This transformation also offers an opportunity for innovation while striving for sustainable production and consumption.

The research using the LCA tool not only provides managers with prospects for improving the existing processes, but also prospects for change. LCA is a powerful tool for guiding technological development, and is particularly important for technologies that still need to be optimised, as LCA can identify process steps and resources that greatly contribute to the overall ecological impact of the life-cycle. Therefore,

the conducted analysis provides managers with a basis for developing the initiative and practical approaches to a successful transition to CE. To obtain the potential environmental benefits of PE waste management, engineers must treat LCA as an opportunity to change their production processes to meet the expectations of the public, which require manufacturing facilities to take the direction of CE.

In conclusion, chemistry, microbiology, and process engineering were combined to develop a highly productive and high-yield process for the conversion of polyethene, a non-degradable polymer, to polyhydroxyalkanoate, a biodegradable polymer. Furthermore, the environmental benefits were assessed through the life-cycle assessment of different PE waste post-processing scenarios. The disclosed process has significantly advanced the chemo-biotechnological recycling of plastic into higher-value products through the demonstration of high cell density and high PHA productivity.

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CRediT authorship contribution statement

Maciej W. Guzik: Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Tomasz Nitkiewicz:** Methodology, Investigation, Data curation, Writing – review & editing. **Magdalena Wojnarowska:** Methodology, Resources. **Mariusz Sołtysik:** Validation, Investigation, Visualization. **Shane T. Kenny:** Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Ramesh P. Babu:** Methodology, Writing – review & editing, Supervision. **Marshall Best:** Resources. **Kevin E. O'Connor:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2021.08.030>.

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